

DustCast

A multi-scale forecasting service for Tunisian rooftop solar production,
with calibrated uncertainty and an evaluated model-selection loop

PESTGM 7.0 CDC Tech Challenge — Track 1

September 2026

What this is. A forecasting service for rooftop PV production at district, governorate and national scale, covering low- and medium-voltage segments. It produces J+0 to J+7, has measured error at J+0 to J+3, quantifies uncertainty, re-trains and re-selects among five competing models each month from observed forecast error, refreshes automatically when new weather arrives, and exposes a documented HTTP API with a working downstream consumer. **What we validated.** Accuracy and interval coverage were measured on real PV production in Alice Springs, Australia. Suitable installation-level Tunisian data were not available to this project, so Tunisian accuracy is unvalidated and the Tunisian fleet is an explicit simulation. **What we found.** Model selection matters more than model sophistication: a boosted model that wins on the array it was developed on loses on arrays it has not seen, so the service selects among models on held-back data rather than committing to one.

1 The service

Tunisia has roughly 500 MW of residential rooftop PV and a further ~ 130 MW on medium- and high-voltage networks. This generation is behind the meter and weather-driven, so the operator carries more reserve than necessary and is exposed to ramps it could have anticipated.

Our modelling puts that in context, *under the stated scenario of §4*. At this penetration there is no duck curve: rooftop PV shaves about 9.9% off midday net load and the overnight demand trough remains the system minimum. The expensive quantity is the evening ramp, where 275 MW of the steepest 781 MW three-hour net-load ramp (35%) is PV falling away rather than demand rising. That share is what a production forecast can anticipate, and it is the operational argument for this service — a scenario output, not a measurement of the Tunisian grid.

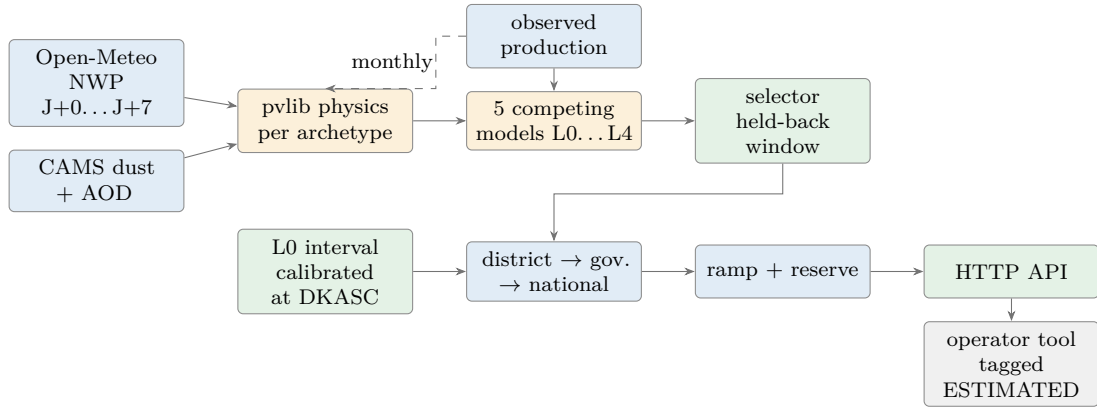


Figure 1: Architecture. Solid path is a forecast request; the dashed path is the learning loop, which refits every candidate monthly on past observations. The operator workflow runs left to right: new weather lands → the service rebuilds → models are scored and one is selected → the forecast is aggregated and banded → a ramp and reserve view is derived → a downstream tool reads it over HTTP, carrying the provenance flags with it.

Capability	Implemented	Evidence in this report
District / governorate / national	266 / 24 / 1	§4, Table 7
LV and MV rooftop segments	3 segments, 630 MW	Table 7
Intra-day and J+1...J+3	produced J+0...J+7	Table 3 (J+0...J+3 only)
Learning from observed error	monthly refit + select	Fig. 2, Table 4
Automatic refresh	on new weather, not a timer	§6
Uncertainty	L0 band + illustrative aggregation	Table 5, Fig. 4
Machine exchange	9 endpoints + consumer	§6

Table 1: Every row is supported by a figure, table or section below.

2 Method

2.1 Physics first, learning on a capacity-normalised residual

A `pvl` model chain computes the power an array should produce from forecast irradiance, temperature and wind given its geometry, capacity and age. Models learn only the remainder. The target is *normalised by AC inverter capacity*:

$$y_t = \frac{P_t^{\text{actual}} - P_t^{\text{expected}}}{P_{\text{nom}}^{\text{AC}}}.$$

The normalisation is not cosmetic. The physics removes the diurnal and seasonal *shape* of production, but a residual expressed in kilowatts still scales with array size. An earlier version of this work trained on kilowatts and over-corrected small arrays by the capacity ratio, producing a positive bias at every hour of the day. Dividing by capacity makes the target dimensionless, so it can be learned on one array and spent on another.

Capacities are reported consistently as *DC module / AC inverter*: array 3 is 4.95 kW DC on a 6.0 kW AC inverter; arrays 16C and 16D are both 1.98 kW DC on 2.5 kW AC. All three are confirmed against the provider’s own listing (§5), so the normalisation denominator is measured rather than assumed. Normalisation and all nMAE denominators use AC.

2.2 Five competing models, and the loop that chooses between them

Model	Fitted from
L0 physics	site geometry only — requires <i>no</i> local production data
L1 + fitted derate	one scalar: mean actual/expected
L2 + diurnal bias	thirteen numbers: mean residual by local hour
L3 + boosted	XGBoost on the normalised residual
L4 + diurnal, then boosted	L2 first, XGBoost on what L2 leaves behind

Table 2: These compete; this is not a progression in which more data eventually promotes the machine-learned model.

At each monthly boundary every candidate is **refitted** on past data only — the boosted models are retrained, not re-ranked — the best performer on a held-back 42-day window that no model was fitted on is selected, and that choice forecasts the following month blind. Updating parameters from observed error is the learning loop; reordering a leaderboard would not be.

3 Results

3.1 Accuracy by horizon

Forecasts are issued from the model run of the stated lead day and are valid for the labelled hour: J+1 means a run issued one day before the valid time. All models are fitted before 2025-03-30 and scored after.

Valid horizon	L0	L1	L2	L3	L4
J+0 (same-day run)	5.57	4.93	3.90	3.10	2.93
J+1	6.67	5.57	4.42	4.23	4.36
J+2	6.70	5.68	4.52	3.72	4.44
J+3	6.74	5.72	4.67	3.78	3.97

Table 3: nMAE (%) of AC capacity, 1,251 daylight hours, DKASC array 3. **Not an independent result** — see the two warnings below.

Error does not increase monotonically from J+1 to J+3 — J+2 scores better than J+1. We report the ordering as observed and draw no conclusion from it: there is no requirement that empirical error rise with lead time, and we ran no test on the horizon-to-horizon differences. J+0 is a same-day model run, not a nowcast; the service has no sub-hourly or sky-imagery input and claims no minute-scale skill.

3.2 What the learning loop did

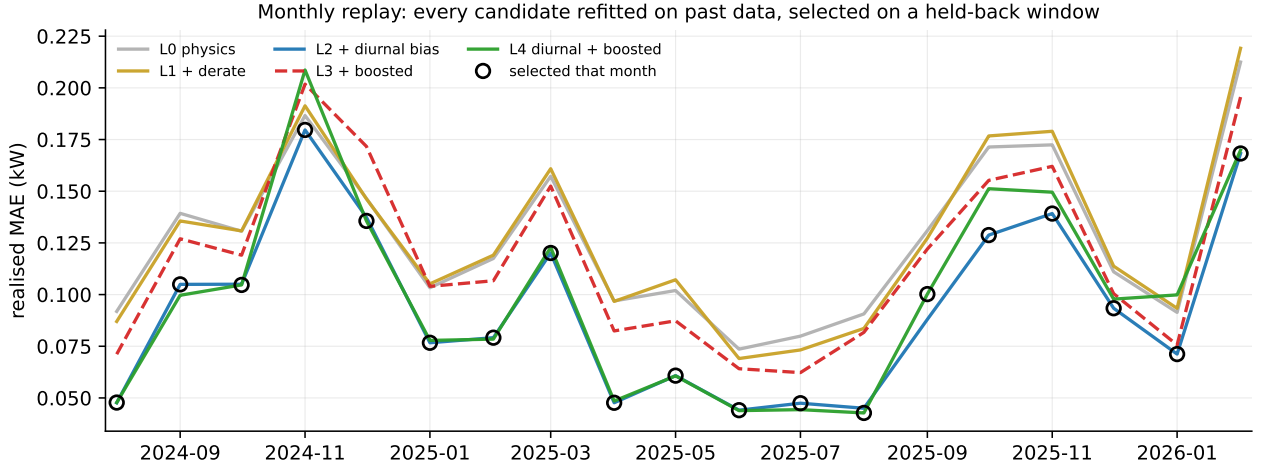


Figure 2: Nineteen months of replay on array 16C. Every candidate is refitted on past data at each boundary; the circled point is the model selected for that month, chosen on a held-back window *before* the month began.

Selection		Model	Pooled nMAE	vs. selector
L2 selected	14 / 19	selector (the service)	3.916 %	—
L4 selected	5 / 19	L2 + diurnal bias	3.897 %	−0.48 %
Switches	7	L4 diurnal + boosted	4.162 %	+5.9 %
Selection was month’s winner	12 / 19	L3 + boosted	4.827 %	+18.9 %
An ML model won the month	8 / 19	L0 physics	5.154 %	+24.0 %

Table 4: Pooled over all 19 months, weighted by hours. The service achieved 3.916 % nMAE. The single best fixed choice in hindsight, L2, achieved 3.897 % — the service was **0.019 percentage points worse, or 0.48 % in relative terms**. It beat every other fixed choice, including both boosted models. A switching policy is not bounded below by the best fixed model in general; here it landed just short of one and well ahead of the rest.

The machine-learned model is competitive rather than decorative: it won outright in 8 of 19 months and was selected in 5. It trails on the pooled figure because its losing months are the heavy ones — an asymmetric error profile, not uniform inferiority.

3.3 Uncertainty, and what it covers

The national layer runs L0 (§4), so its interval must be built for L0. We calibrate a split-conformal band on arrays that *have* measured production, as a fraction of AC capacity, binned by sun elevation

and clear-sky index — both known at forecast time. Half-width at 90 % nominal is 0.146 of AC capacity overall, widening under cloud (0.161–0.168 for clear-sky index 0.4–0.7) and narrowing in clear high sun (0.101 above 45°).

Array	azimuth	hours	PICP	PINAW
DKASC 3	0°	1,272	94.1 %	0.259
DKASC 16C	90°	3,269	92.1 %	0.245
DKASC 16D	280°	1,347	86.4 %	0.240

Table 5: Out-of-sample coverage of the L0 band, calibrated on 11,246 daylight hours before 2025-03-30 and scored after. Nominal 90 %. Coverage and sharpness are reported together: either alone is gameable, since a band from zero to infinity has perfect coverage and no value.

Aggregating those site bands to a national figure is where the evidence runs out, and we label it accordingly. We combine regional widths with $\sqrt{(1 + (n - 1)\rho)/n}$, which recovers the plain sum at $\rho = 1$ and the $\sqrt{n}$ reduction at $\rho = 0$, taking ρ as the mean pairwise correlation of the forecast clear-sky index across the 24 governorates ($\rho = 0.435$ in the snapshot this report quotes throughout; it moves with the weather window). At the forecast peak of that snapshot the aggregated range is 67.8 MW wide — **± 33.9 MW** on 452.6 MW — against a summed width of 100.1 MW if errors were assumed perfectly correlated and 20.4 MW if independent.

This is an illustrative sensitivity calculation, and the national display is labelled as a range rather than a 90 % interval. Three assumptions in it are not satisfied. The shrinkage factor assumes equal regional error scales and one common correlation, whereas the regions are population-weighted and carry very unequal capacity and width. It treats conformal half-widths as though they were standard deviations, which they are not. And ρ is measured on the forecast clear-sky *index* as a proxy for forecast-*error* correlation — a substitution our own earlier measurement showed can differ substantially (clear-sky index correlation 0.444 against actual forecast-error correlation 0.089 on the same data, and 0.607 in summer against 0.022 in winter).

Table 5 establishes out-of-sample coverage on three arrays at *one* Australian facility — not three independent sites, and not a spatial test. National coverage over Tunisia is not established. The caveat travels with the number: it labels the dashboard and the chart, and every API payload carrying the range names its bounds `low_illustrative` and `high_illustrative`.

3.4 Generalisation across arrays

Having revised models, baselines and calibration while inspecting one window, we froze the pipeline and evaluated it on array 16C — a different array, east facing rather than north, 1.98 kW DC rather than 4.95, over a later period.

Stage	Configuration	Skill vs. L2	95 % CI
(i) initial frozen evaluation	as frozen	−46.8 %	[−53.7, −41.0]
(ii) after correcting a degenerate ageing fit	post-hoc	−48.3 %	[−55.3, −42.2]
(iii) capacity-normalised target	post-hoc	−23.1 %	
(iv) + site bias adaptation	post-hoc	−15.6 %	
fresh array 16D (west), 2 training azimuths	post-hoc	−9.4 %	[−12.1, −7.0]

Table 6: Only (i) is a frozen-protocol result. Everything below it was introduced *after* inspecting that result — the ageing fit was degenerate and was repaired, the unit error was found and fixed, and the site-bias variant was added to give the boosted model the same site adaptation the L2 baseline receives. Refitting those on earlier dates does not undo the fact that the changes were test-informed. They are diagnostics of *why* the model failed to transfer, not independent confirmations that it now does.

The pattern is consistent: the boosted model wins on the array it was developed on (Table 3) and loses on arrays it has not seen. That is the signature of overfitting to a site. It is also why the service

selects among models rather than committing to one, and why the selector — not the boosted model — is the contribution we claim.

3.5 Operational implications, under a stated scenario

The forecast is only useful if it changes a decision. To show what it would change, we combine the simulated fleet with an assumed evening-peaking demand shape peaking at 4,600 MW. **Everything in this subsection is an output of those assumptions, not a measurement of the Tunisian grid;** the assumptions are listed in Table 7. We report it because the shape of the answer — where in the day rooftop PV actually matters to an operator — is robust to the exact numbers in a way the magnitudes are not.

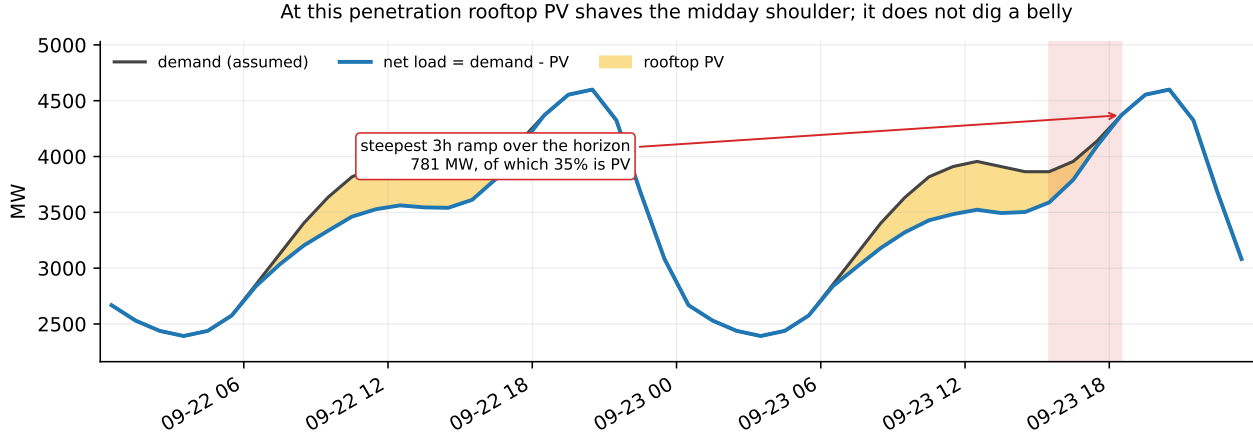


Figure 3: Two modelled days of national net load, under the assumed demand shape and simulated fleet of §4. **These are scenario outputs, not measurements of the Tunisian grid.**

Under that scenario, rooftop PV removes about 9.9 % of midday net load without creating a midday minimum — the system trough remains overnight — and PV falling away accounts for 275 MW of the steepest 781 MW three-hour evening ramp. If the real fleet and demand resemble these assumptions, roughly a third of the evening ramp is the component a production forecast can anticipate.

A reserve-procurement simulation — a *separate, earlier* experiment, run on its own assumptions of a 500 MW fleet scaled from a single array against a 4,000 MW demand peak, not the 630 MW fleet and 4,600 MW peak used above — gave a 3.6 % reduction in forecast-sensitive decision cost for a boosted forecast over L2, but at *lower achieved reliability* (unserved energy 65.9 against 36.0 MWh), and the sign of the saving reverses across plausible cost assumptions (range −9.9 % to +6.4 % over twenty cases). We therefore report it as inconclusive. Separately, using a fixed 90 % display interval directly as a reserve policy performed poorly (−41.1 % against the baseline, −46.4 % against the same model’s scalar band); the narrow conclusion is about that policy, not about coverage-calibrated bands in general.

4 Deployment assumptions and sources

Quantity	Value used	Basis and status
Residential rooftop	500 MW (LV)	STEG statement, end-June 2026. The Ministry has separately cited “~400 MW installed vs. 70 MW operational”; we flag the conflict rather than resolving it
Commercial / services	78 MW (LV-MV)	our split of the ~130 MW MV/HV figure
Industrial	52 MW (MV)	our split , as above
Capacity by governorate	population-weighted	assumption ; real PROSOL uptake is unlikely to be proportional to population
Aggregation units	266, from delegation counts	prototype units — see below
Tilt / azimuth / size mix	per-segment distributions	assumption ; no registry was available to fit them
Demand shape and peak	4,600 MW evening-peaking	assumption for the scenario in §3.5

Table 7: Every entry in bold is ours, not a published figure.

On aggregation units. We subdivide each governorate into as many units as it has delegations, and treat those as prototype aggregation units. We have *not* established that they correspond to the operational units STEG would want, which are more likely defined by network topology — HV/MV substations and MV feeders — than by administrative boundaries. The intended mapping is a lookup from each installation’s connection point to its feeder and substation, replacing the administrative split without changing the aggregation code, which sums whatever units it is given. Unit *names* are not modelled.

On weather resolution. All units within a governorate currently share one weather series. This is a modelling simplification we chose to keep the service to 24 grid requests, *not* a limit imposed by the data: an ~11 km forecast grid resolves variation well inside a Tunisian governorate, and finer sampling is a matter of more requests. CAMS dust at ~40 km is the coarser constraint.

On data availability. Suitable installation-level data were not available to this project: we found no accessible per-site registry or generation time series for Tunisian rooftop PV. This is a statement about what we could obtain, not a claim that no such records exist — ANME guidance describes commissioning records and inverter production histories, which would materially change what is possible here if accessible.

5 Reproducibility

Every number and figure is regenerated from scripts in the repository; no figure is drawn from retyped values.

Dataset	Identity	Use
DKASC array 3	70-Site_DKA-M5_A-Phase, 4.95/6.0 kW, az. 0°	training, horizons
DKASC 16C	1.98/2.5 kW, az. 90°	frozen evaluation, replay
DKASC 16D	1.98/2.5 kW, az. 280° (<i>fitted, unverified</i>)	fresh evaluation
Open-Meteo forecast	hourly, J+0...J+7	live weather
Open-Meteo historical forecast	archived runs, leads 1,2,3,5,7	horizon evaluation
Open-Meteo air quality	CAMS, domains=cams_global	dust, AOD

Table 8: Access as used by this prototype: no API key was required and no fee was paid. Open-Meteo’s hosted free tier is for **non-commercial** use and is rate limited, so an operational deployment would need their commercial tier or a self-hosted equivalent. Alice Springs production spans 2008–2026 at 5-minute resolution, resampled hourly with bins relabelled to their midpoint before solar geometry is evaluated.

Boundaries. Horizon evaluation fits before 2025-03-30 and scores after (1,251 hours). The replay

runs 2024-08 to 2026-02, refitting monthly, selecting on the 42 days before each boundary and scoring the month after. Interval calibration uses 11,246 daylight hours before 2025-03-30 across three arrays; coverage is scored after. Daylight is sun elevation $> 5^\circ$; all nMAE denominators are AC inverter capacity.

Commands. Each script prints its own result table and writes a CSV to `data/artifacts/`.

Script	Produces
<code>step16_replay.py</code>	the learning loop, Fig. 2 and Table 4
<code>step17_horizons.py</code>	accuracy by horizon, Table 3
<code>step18_district_national.py</code>	district $\rightarrow$ governorate $\rightarrow$ national
<code>step21_national_uncertainty.py</code>	interval calibration and aggregation, Table 5
<code>step19_consumer.py</code>	API and downstream consumer, end to end
<code>step14_frozen_holdout.py</code>	the frozen evaluation, stage (<i>i</i>)
<code>step15_multiarray.py</code>	normalisation and multi-array diagnostics
<code>step20_report_figures.py</code>	every figure in this report

6 The operator interface

A forecast is only operationally useful if another system can consume it without a human in between. The service therefore exposes the forecast, its uncertainty and its evaluation record over HTTP, and we ship a consumer that exercises the whole path — connect, check provenance, read the series, observe the refresh behaviour, derive ramp and reserve quantities, and write a handoff file.

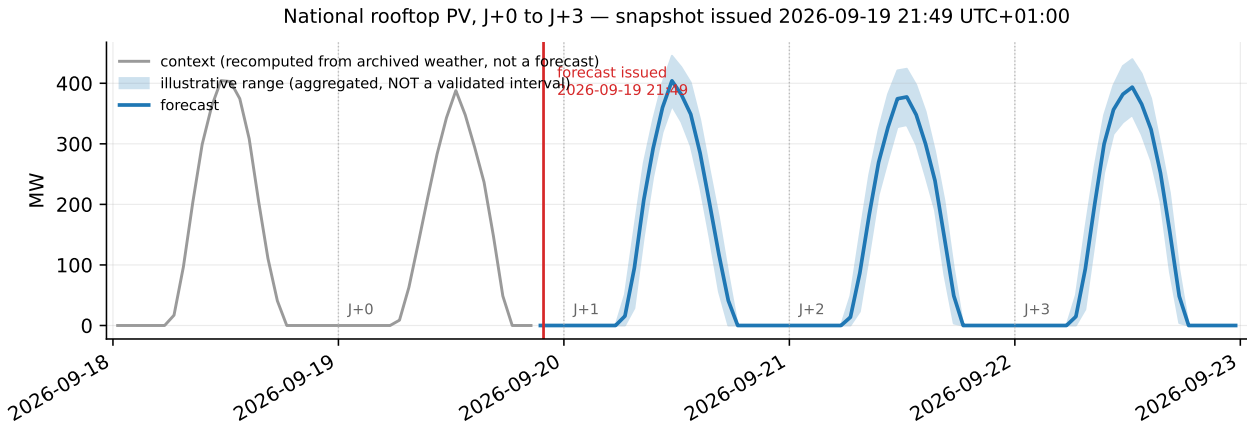


Figure 4: The product, from the snapshot issued 2026-09-19 21:49 that every number in this report is drawn from. The red line is the **issue time**, and J+0 is counted from it. Hours to its left are **context**: values recomputed from archived weather for elapsed hours, drawn in grey and *not* forecasts of those hours — a freshly generated estimate for an hour that has already passed is not an operational forecast of it. The shaded band on the forecast side is the **illustrative range** of §3.3, not a validated 90% interval. The snapshot holds 69 context hours and 146 forecast hours.

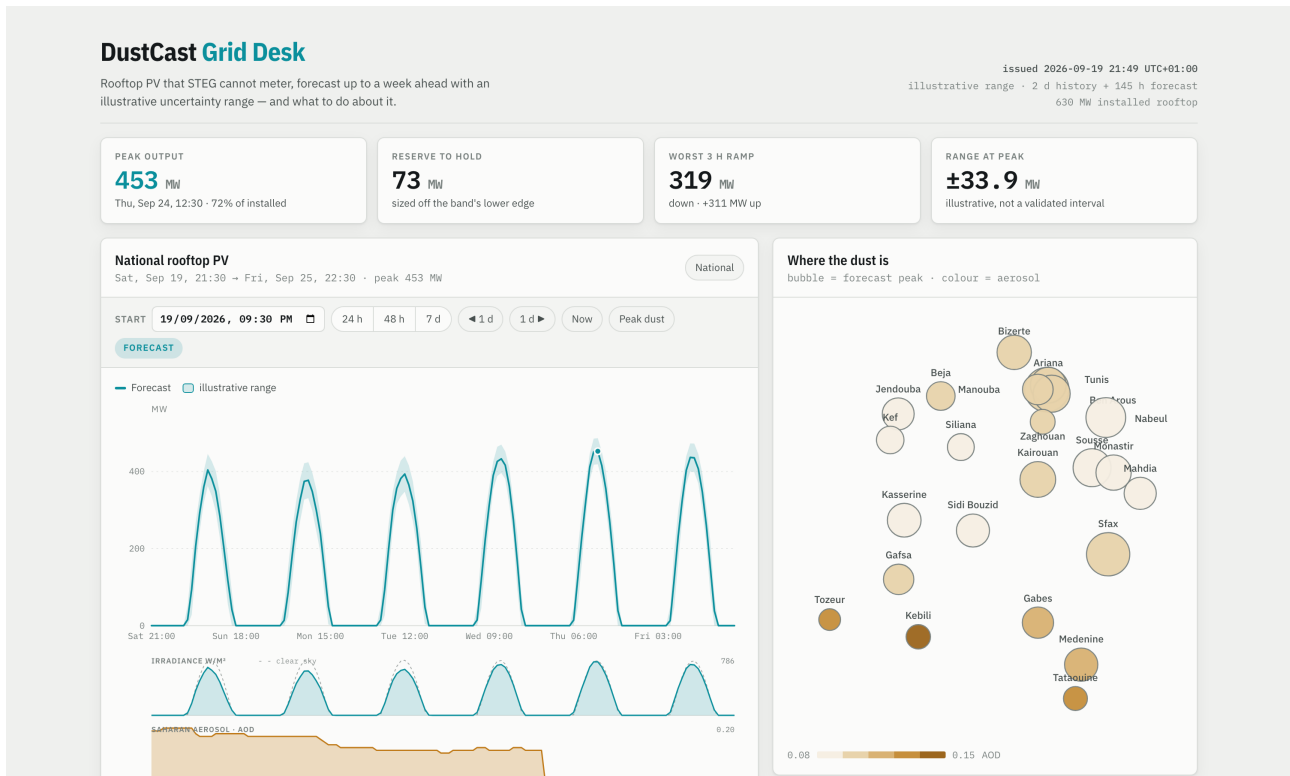


Figure 5: The operator dashboard, rendered from the same snapshot (§5 regenerates it). It reports 630 MW installed and an issue time of 2026-09-19 21:49, and its “range at peak” tile reads ± 33.9 MW — the same figures as §3.3 and the API response below, which is the point of building all four from one snapshot. The band is labelled an illustrative range here too. The map sizes each governorate by forecast peak and colours it by aerosol optical depth.

Endpoint	Returns
GET /health	freshness, horizon length
GET /meta	scales, segments, capacity, provenance flags
GET /forecast/national	national MW series
GET /forecast/governorate/{key}	one governorate
GET /forecast/governorates	all 24, in the shape a map wants
GET /forecast/district/{key}	one unit, with its segment mix
GET /models/performance	measured nMAE by horizon
GET /models/selection	what the loop chose, month by month
POST /refresh	force a rebuild

Table 9: The evaluation record is served alongside the forecast, so a consumer can see what the numbers are worth rather than having to trust them.

```
$ curl -s localhost:8000/forecast/national
{ "generated_at": "2026-09-19T21:49:57+01:00",
  "units": "MW (AC)", "model": "L0 physics",
  "fleet": "SIMULATED -- no installation registry available to this project",
  "evaluated_horizons": ["J+0", "J+1", "J+2", "J+3"],
  "uncertainty": {
    "status": "ILLUSTRATIVE -- not a validated interval",
    "caveat": "ILLUSTRATIVE RANGE, NOT A VALIDATED 90% INTERV...",
    "spatial_correlation_rho": 0.4348 },
  "series": [ { "time": "2026-09-24T12:30:00+01:00",
    "mw": 452.6187,
    "low_illustrative": 418.7306,
```

```
"high_illustrative": 486.5068 }, ... ] }
```

A response captured from the running service by `step22_snapshot.py`, not transcribed by hand. Its `generated_at` is when that *response* was built and is not the forecast issue time.

Provenance travels with the numbers: a downstream tool that silently treated a simulated national total as a metered one is the failure mode this interface exists to prevent, and the sample consumer reads those flags and tags its output **ESTIMATED**. Freshness is explicit, and a rebuild is triggered by new weather arriving rather than by a timer.

7 Limitations

- **Tunisian accuracy is unvalidated.** The 5.57 % and 6.67 % figures were measured at DKASC in Australia. No Tunisian production series was available to score against.
- **National interval coverage is unvalidated**, for the same reason; only site coverage (Table 5) is measured, and ρ is estimated.
- **The fleet is simulated**, including capacity allocation and segment mix (Table 7).
- **Aggregation units are prototypes**, not verified STEG operational units.
- **The boosted model does not improve accuracy on unseen arrays**, and is used only when selected.
- **Scenario results in §3.5** depend on an assumed demand shape and fleet; they are not measurements of the Tunisian grid.
- **Soiling has no ground truth** and is presented as a relative index and maintenance trigger, never a measured loss.
- **J+4 to J+7 are produced but not evaluated.**
- **The dashboard is still fed by an earlier single-segment pipeline** (500 MW residential only) and has not yet been repointed at the service described here, so its figures do not match this report. Repointing it is the first task before the demo.

8 Conclusion

DustCast is a working multi-scale rooftop PV forecasting service: district to national, low- and medium-voltage segments, J+0 to J+7 produced with measured error to J+3, per-site conformal intervals with measured coverage and an explicitly illustrative national aggregation, monthly retraining and selection from observed forecast error, automatic refresh, and a documented machine interface with a working consumer.

Its measured contribution is the selection loop. Five models compete; the service achieved 3.916 % nMAE against 3.897 % for the best fixed choice identifiable only in hindsight, and beat every other fixed choice by 6–24 %. That matters because the ranking is not stable: the boosted model wins on the array it was developed on and loses on arrays it has not seen, so a service that commits to one model in advance is making a bet the evidence does not support.

What that does *not* show is an accuracy improvement over L2: the service finished marginally behind it. The aggregation assumptions, the transfer behaviour and the selection policy all still require validation. Access to Tunisian installation records and metered production would make that validation possible — it would not by itself establish reliable performance.

Repository and reproduction

Repository
Dashboard

«< INSERT GITHUB URL BEFORE SUBMISSION »>
«< INSERT DASHBOARD URL BEFORE SUBMISSION »>

```
git clone <repository-url> && cd PESTGM
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
```

```
# Production data (~880 MB) is not committed. Exact provider exports:
B=https://solarcentre.spinifexvalley.com.au/export
curl -C - -o data/raw/dkasc_site03_5min.csv $B/70-Site_DKA-M5_A-Phase.csv
curl -C - -o data/raw/fleet/dka_16c_east.csv $B/105-Site_DKA-M1_C-Phase.csv
```

```
curl -C - -o data/raw/fleet/dka_16d_west.csv $B/81-Site_DKA-M2_A-Phase.csv
```

```
.venv/bin/python scripts/step16_replay.py          # learning loop (Fig 2, Tab 4)
.venv/bin/python scripts/step17_horizons.py        # horizons (Tab 3)
.venv/bin/python scripts/step21_national_uncertainty.py # intervals (Tab 5)
.venv/bin/python scripts/step18_district_national.py # aggregation
.venv/bin/python scripts/step20_report_figures.py   # every figure
.venv/bin/uvicorn dustcast.api:app --port 8000      # the service
.venv/bin/python scripts/step19_consumer.py        # API, end to end
```

Array	Provider source	Bytes	SHA-256 (first 16)
3	70, Site_DKA-M5_A-Phase	313,051,557	0cb0746fe5abda4e
16C	105, Site_DKA-M1_C-Phase	285,663,460	2b155248911195b2
16D	81, Site_DKA-M2_A-Phase	284,948,839	b829d6ce08612026

Provenance, resolved. An earlier draft reported 16C and 16D as unidentified. The identifiers were recovered from the provider’s download page, whose export links are written into `data-href` attributes by script, and all three remote `Content-Length` values match the local files exactly. The provider labels source 105 “BP Solar, 2.0 kW, poly-Si, Fixed, 2008, **East**” and source 81 the same but **West**, which confirms the capacity and orientation of both. Array 16 is a deliberate orientation study (16A north, 16B flat, 16C east, 16D west). 16D’s azimuth had been recovered by fitting before that label was found — best fit 280°, on clear high-sun hours in 2015–2020, outside every evaluation window — and the independent fit agrees with the provider’s label, which is a check on the fitting method rather than a coincidence to conceal. Nothing about these two arrays is now provisional.

Sources

- [1] STEG, statement on installed residential rooftop and MV/HV connected PV capacity, end-June 2026 — the 500 MW and ~130 MW figures in Table 7. The Ministry separately cited “~400 MW installed vs. 70 MW operational” for low-voltage PV (Dec. 2025), which appears to conflate pipeline with commissioned capacity; we flag the conflict.
- [2] IRENA, *Renewable Capacity Statistics*, 2026 — national PV fleet ~895 MW, for context.
- [3] ANME, guidance describing PROSOL installation commissioning records and inverter production histories — evidence that installation-level records are described as existing. We were unable to obtain an accessible dataset (§4).
- [4] Desert Knowledge Australia Solar Centre (DKASC), Alice Springs — per-array 5-minute production exports, <https://dkasolarcentre.com.au/download>; per-array metadata at <https://dkasolarcentre.com.au/source/alice-springs/>. All three arrays used here are at this one facility.
- [5] Open-Meteo, Weather Forecast / Historical Forecast / Air Quality APIs, <https://open-meteo.com/en/docs>, <https://open-meteo.com/en/docs/historical-forecast-api>, <https://open-meteo.com/en/docs/air-quality-api>. Terms: <https://open-meteo.com/en/terms> — the free tier is non-commercial and rate limited. Dust and AOD are CAMS-derived; `domains=cams_global` is required for Tunisian coverage.
- [6] Copernicus Atmosphere Monitoring Service (CAMS), global atmospheric composition forecasts.
- [7] `pvlib python`, <https://pvlib-python.readthedocs.io> — Ineichen–Perez clear-sky, Hay–Davies transposition, PVWatts DC and inverter models, SAPM cell temperature.
- [8] AEMO (Australia) and Sheffield Solar / Open Climate Fix (UK) — the sample-to-region upscaling method adopted in §4. Their samples are metered; ours is simulated.
- [9] V. Vovk, A. Gammerman and G. Shafer, *Algorithmic Learning in a Random World* (Springer, 2005) — the finite-sample conformal quantile $\lceil (n+1)\alpha \rceil / n$ used for the L0 band in Table 5.
- [10] L. Micheli et al., on the impact of a Saharan dust outbreak on photovoltaic performance, *Sustainable Energy Technologies and Assessments* **62** (2024) — a March 2022 outbreak that roughly halved the Spanish PV fleet’s capacity factor for over two weeks, with an ~80 % drop on the peak day.
- [11] *Atmospheric Chemistry and Physics* **25**, 997 (2025), <https://acp.copernicus.org/articles/25/997/2025/> — a **specific modelled March 2021 West African dust event** in which surface solar irradiance was reduced by ~18 %; that event, not a dry-season average. With [10] it establishes mechanism and order of magnitude for dust impact, **not** Tunisian values.
- [12] Decree 2016-1123 — Tunisian net metering, capping surplus sales to STEG at 30 % of annual production; relevant to self-consumption sizing in the fleet assumptions.